

SmartUSBHub

Communication Protocol

Applies to: SmartUSBHub Product Family

Released v1.0

www.mixedsignallab.com

Contents

Document Navigation	2
Command Set	2
Frame format	3
V1 frame: one data byte	3
V2 frame: two data bytes	3
V3 frame: variable-length data	3
Command Examples	4
Control Channel Power Switch (CMD: 0x01)	5
Query Channel Power State (CMD: 0x00)	7
Control Channel USB Data Line Switch (CMD: 0x05)	9
Query Channel USB Data Line State (CMD: 0x08)	11
Control Power in Interlock Mode (CMD: 0x02)	13
Query Channel Voltage (CMD: 0x03)	14
Query Channel Current (CMD: 0x04)	15
Set Button Control Permission (CMD: 0x09)	17
Query Button Control Permission (CMD: 0x0A)	17
Set Channel Power Default State (CMD: 0x0B)	18
Query Channel Power Default State (CMD: 0x0C)	21
Set Default Data Connectivity State (CMD: 0x0D)	23
Query Default Data Connectivity State (CMD: 0x0E)	26
Set Power-Loss State Persistence (CMD: 0x0F)	28
Query Power-Loss State Persistence (CMD: 0x10)	29
Set Device Address (CMD: 0x11)	29
Get Device Address (CMD: 0x12)	31
Set Operation Mode (CMD: 0x06)	32
Query Operation Mode (CMD: 0x07)	33
Reset Factory Settings (CMD: 0xFC)	33
Query Firmware Version (CMD: 0xFD)	33
Query Hardware Version (CMD: 0xFE)	34

[Product Documentation](#) • [Python SDK](#)

This document describes the shared serial settings, protocol frame format, complete command set, and command examples for the SmartUSBHub family. Channel counts and optional capabilities vary by product; the channel values and frames below primarily use a four-channel model as the example.

Document Navigation

- [Command set](#)
- [Frame format](#)
- [Command examples](#)
- [Product documentation](#)
- [Python SDK](#)

Command Set

The device uses a simple request-response communication protocol for control. The available commands are:

Function	CMD	Description
Control channel power	0x01	Control the VBUS power state of a specified channel
Query channel power state	0x00	Query whether a channel' s VBUS power is on
Control power in interlock mode	0x02	Enable power on only one channel at a time
Query voltage	0x03	Query the channel' s VBUS voltage (unit: mV)
Query current	0x04	Query the channel' s current (unit: mA)
Control USB data lines	0x05	Control whether D+/D- are connected
Query USB data line status	0x08	Query the current on/off status of D+/D-
Set button control permission	0x09	Enable or disable the physical button control
Query button control permission	0x0A	Query whether button control is enabled
Set power default state	0x0B	Configure the default on/off state of VBUS after power-up
Query power default state	0x0C	Query the channel' s power default state
Set data default state	0x0D	Configure the default on/off state of D+/D- after power-up
Query data default state	0x0E	Query the default state of D+/D-
Set power-loss state Persistence	0x0F	Enable/disable remembering channel state on power loss
Query power-loss Persistence state	0x10	Query if power-loss state Persistence is enabled
Set Device Address	0x11	Set the device address, used to identify and distinguish hubs in multi-hub setups
Get Device Address	0x12	Get the device address, used to identify and distinguish hubs in multi-hub setups
Set operation mode	0x06	Switch Normal/Interlock mode
Query operation mode	0x07	Query the current operating mode
Restore factory settings	0xFC	Restore factory settings
Query firmware version	0xFD	Query the device firmware version
Query hardware version	0xFE	Query the current hardware version

Frame format

SmartUSBHub uses V1, V2, and V3 frames on the same serial link. Numbers in the layouts below are bit offsets; each regular field occupies 8 bits (one byte).

V1 frame: one data byte

Offset	Field	Size	Description
0	SOF1	1	Fixed 0x55
1	SOF2	1	Fixed 0x5A
2	CMD	1	Command code
3	CHANNEL	1	Channel bitmask; bit 0 is channel 1
4	VALUE	1	One-byte command argument or status
5	SUM8	1	(CMD + CHANNEL + VALUE) & 0xFF

V1 is always 6 bytes and carries most set, query, and acknowledgement messages.

V2 frame: two data bytes

Offset	Field	Size	Description
0–1	SOF	2	Fixed 55 5A
2	CMD	1	Command code
3	CHANNEL	1	Channel bitmask
4	VALUE0	1	High byte of voltage/current, or the enable field
5	VALUE1	1	Low byte of voltage/current, or the status field
6	SUM8	1	(CMD + CHANNEL + VALUE0 + VALUE1) & 0xFF

V2 is always 7 bytes. It currently carries voltage/current replies and default power/data-line states that need two data bytes. Parse a 16-bit measurement as $VALUE0 \ll 8 \mid VALUE1$.

V3 frame: variable-length data

Offset	Field	Size	Description
0–3	MAGIC	4	Fixed 55 AB CD EF
4	CMD	1	Command code
5	FLAGS	1	bit 0=STREAM; set for unsolicited stream notifications
6–7	LENGTH	2	Payload length, little-endian: LEN_L, LEN_H
8–9	CRC16	2	Little-endian: CRC_L, CRC_H
10...	PAYLOAD	LENGTH	Command-specific variable data

The total V3 frame size is $10 + LENGTH$ bytes. The current maximum is 64 bytes, leaving up to 54 payload bytes. CRC16 uses polynomial 0x8005 and initial value 0xFFFF; zero both CRC field bytes before calculating across the complete frame from MAGIC through the end of PAYLOAD. V3 carries batch measurements, measurement streams, channel names, device aliases, debug logs, and other variable-length data.

**NOTE: The SDK selects and parses the frame version automatically**

Normal API calls do not need to specify V1, V2, or V3.

Command Examples

**NOTE**

- *All data values below are in hexadecimal.
- *Channel values use the [Channel_e](#) enum, not the channel number. In control and query commands, bitwise OR can be used for batch control and query; if the device is in interlock mode, only the [interlock control command \[0x02\]](#) can be used for control; bitwise combination is ineffective.
- *Frames with an incorrect format or unsupported commands will not receive a response.
- *When a button is pressed, a [status report \[0x00\] command](#) of that channel's current state will be sent.
- *The provided package includes a UART assistant tool (uart_tool) for quick start.
- *A Python control library [smartusbhub](#) is available.
- *In this document, USB downstream port is referred to simply as a channel.

Channel Enum Values (Channel_e):

```
typedef enum
{
    CH1 = 0x01,
    CH2 = 0x02,
    CH3 = 0x04,
    CH4 = 0x08,
    CH5 = 0x10,
    CH6 = 0x20,
    CH7 = 0x40,
}Channel_e;
```

The available channels depend on the product model: four-channel products use CH1 through CH4, while seven-channel products use CH1 through CH7.

**NOTE****Device Communication Parameters:**

- *Baud Rate: 115200 – 921600 (Default: [115200](#))
- *Data Bits: [8](#)
- *Parity: [None](#)
- *Stop Bits: [1](#)

Device Name:

- *Windows: [COMx](#)
- *Linux: [/dev/ttyACMx](#)
- *macOS: [/dev/cu.usbmodemx](#)

Control Channel Power Switch (CMD: 0x01)



NOTE: If the device is in interlock mode, the normal control command (CMD: 0x01) is ineffective; you must use the interlock control command (CMD: 0x02) instead

If a control command is sent in interlock mode, an “invalid command” frame will be returned:

Invalid Command:

55 5A 01 FF FF FF

Data Field Description

data[0]	data[1]
Channel ID	Channel Power State
Channel_e	0 (OFF), 1 (ON)

Turn On Channel 1 Power

Command:

55 5A 01 01 01 03

Response:

55 5A 01 01 01 03

Turn Off Channel 1 Power

Command:

55 5A 01 01 00 02

Response:

55 5A 01 01 00 02

Turn On Channel 2 Power

Command:

55 5A 01 02 01 04

Response:

55 5A 01 02 01 04

Turn Off Channel 2 Power

Command:

55 5A 01 02 00 03

Response:

55 5A 01 02 00 03

Turn On Channel 3 Power

Command:

55 5A 01 04 01 06

Response:

55 5A 01 04 01 06

Turn Off Channel 3 Power

Command:

55 5A 01 04 00 05

Response:

55 5A 01 04 00 05

Turn On Channel 4 Power

Command:

55 5A 01 08 01 0A

Response:

55 5A 01 08 01 0A

Turn Off Channel 4 Power

Command:

55 5A 01 08 00 09

Response:

55 5A 01 08 00 09

Combined Control Example

Turn On Channel 1 and Channel 3 Power

Command:

55 5A 01 05 01 07

Response:

55 5A 01 05 01 07

Turn Off Channel 1 and Channel 3 Power

Command:

```
55 5A 01 05 00 06
```

Response:

```
55 5A 01 05 00 06
```

Turn On All Channels

Command:

```
55 5A 01 0F 01 11
```

Response:

```
55 5A 01 0F 01 11
```

Turn Off All Channels

Command:

```
55 5A 01 0F 00 10
```

Response:

```
55 5A 01 0F 00 10
```

Query Channel Power State (CMD: 0x00)

Data Field Description

data[0]	data[1]
Channel ID	Channel Power State
Channel_e	0 (OFF), 1 (ON)

Query Channel 1 Power State

Command:

```
55 5A 00 01 00 01
```

Response:**If channel 1 power is OFF:**

```
55 5A 00 01 00 01
```

If channel 1 power is ON:

```
55 5A 00 01 01 02
```


Query Channel 2 Power State

Command:

```
55 5A 00 02 00 02
```

Response:

If channel 2 power is OFF:

```
55 5A 00 02 00 02
```

If channel 2 power is ON:

```
55 5A 00 02 01 03
```

Query Channel 3 Power State

Command:

```
55 5A 00 04 00 04
```

Response:

If channel 3 power is OFF:

```
55 5A 00 04 00 04
```

If channel 3 power is ON:

```
55 5A 00 04 01 05
```

Query Channel 4 Power State

Command:

```
55 5A 00 08 00 08
```

Response:

If channel 4 power is OFF:

```
55 5A 00 08 00 08
```

If channel 4 power is ON:

```
55 5A 00 08 01 09
```

Combined Control Example

Query All Channels' Power States

Command:

```
55 5A 00 0F 00 0F
```

Response:

```
55 5A 00 01 01 02
55 5A 00 02 00 02
55 5A 00 04 00 04
55 5A 00 08 01 09
```

- **Channel 1 Power: ON**
- **Channel 2 Power: OFF**
- **Channel 3 Power: OFF**
- **Channel 4 Power: ON**

Control Channel USB Data Line Switch (CMD: 0x05)

The device can control the connectivity of the D+ and D– differential pair for a specified channel, allowing scenarios where power is maintained but data is disconnected.

**NOTE**

By default, each channel's USB data (D+/D–) lines are connected unless the user sends a control command (CMD: 0x05) to manually disconnect them.

If power-loss state Persistence is enabled, the data line on/off state will be remembered.

Data Field Description

data[0]	data[1]
Channel ID	Channel Data Line State
Channel_e	0 (OFF), 1 (ON)

Connect Channel 1 Data Lines**Command:**

```
55 5A 05 01 01 07
```

Response:

```
55 5A 05 01 01 07
```

Disconnect Channel 1 Data Lines**Command:**

```
55 5A 05 01 00 06
```

Response:

```
55 5A 05 01 00 06
```

Connect Channel 2 Data Lines**Command:**

55 5A 05 02 01 08

Response:

55 5A 05 02 01 08

Disconnect Channel 2 Data Lines

Command:

55 5A 05 02 00 07

Response:

55 5A 05 02 00 07

Connect Channel 3 Data Lines

Command:

55 5A 05 04 01 0A

Response:

55 5A 05 04 01 0A

Disconnect Channel 3 Data Lines

Command:

55 5A 05 04 00 09

Response:

55 5A 05 04 00 09

Connect Channel 4 Data Lines

Command:

55 5A 05 08 01 0E

Response:

55 5A 05 08 01 0E

Disconnect Channel 4 Data Lines

Command:

55 5A 05 08 00 0D

Response:

```
55 5A 05 08 00 0D
```

Combined Control Example

Connect Data Lines on All Channels

Command:

```
55 5A 05 0F 01 15
```

Response:

```
55 5A 05 0F 01 15
```

Disconnect Data Lines on All Channels

Command:

```
55 5A 05 0F 00 14
```

Response:

```
55 5A 05 0F 00 14
```

Query Channel USB Data Line State (CMD: 0x08)

Data Field Description

data[0]	data[1]
Channel ID	Channel Data Line State
Channel_e	0 (OFF), 1 (ON)

Query Channel 1 Data Line State

Command:

```
55 5A 08 01 00 09
```

Response:

If channel 1 data is DISCONNECTED:

```
55 5A 08 01 00 09
```

If channel 1 data is CONNECTED:

```
55 5A 08 01 01 0A
```

Query Channel 2 Data Line State

Command:

```
55 5A 08 02 00 0A
```

Response:

If channel 2 data is DISCONNECTED:

```
55 5A 08 02 00 0A
```

If channel 2 data is CONNECTED:

```
55 5A 08 02 01 0B
```

Query Channel 3 Data Line State

Command:

```
55 5A 08 04 00 0C
```

Response:

If channel 3 data is DISCONNECTED:

```
55 5A 08 04 00 0C
```

If channel 3 data is CONNECTED:

```
55 5A 08 04 01 0D
```

Query Channel 4 Data Line State

Command:

```
55 5A 08 08 00 10
```

Response:

If channel 4 data is DISCONNECTED:

```
55 5A 08 08 00 10
```

If channel 4 data is CONNECTED:

```
55 5A 08 08 01 11
```

Combined Control Example

Combined Control Example: Query All Channels' Data Line States

Command:

```
55 5A 08 0F 00 17
```

Response:

```
55 5A 08 01 01 0A  
55 5A 08 02 01 0B  
55 5A 08 04 01 0D  
55 5A 08 08 01 11
```

- **Channel 1 Data: CONNECTED**
- **Channel 2 Data: CONNECTED**
- **Channel 3 Data: CONNECTED**
- **Channel 4 Data: CONNECTED**

Control Power in Interlock Mode (CMD: 0x02)



CAUTION: In interlock mode, use only this command to select the powered channel; the regular power command (CMD: 0x01) cannot control channels

Turning a channel's power off also disconnects its data line. CMD: 0x02 itself controls power; use the data-line command (CMD: 0x05) when the selected channel's USB2 data line also needs to be controlled.

Control Truth Table:

Channel 1	Channel 2	Channel 3	Channel 4
Closed	Closed	Closed	Closed
Open	Closed	Closed	Closed
Closed	Open	Closed	Closed
Closed	Closed	Open	Closed
Closed	Closed	Closed	Open

Data Field Description

data[0]	data[1]
Channel ID	Channel Power State
Channel_e	0 (OFF), 1 (ON)

Turn On Channel 1 Power (others off)

Command:

```
55 5A 02 01 01 04
```

Response:

```
55 5A 02 01 01 04
```

Turn On Channel 2 Power (others off)

Command:

```
55 5A 02 02 01 05
```

Response:

```
55 5A 02 02 01 05
```

Turn On Channel 3 Power (others off)**Command:**

55 5A 02 04 01 07

Response:

55 5A 02 04 01 07

Turn On Channel 4 Power (others off)**Command:**

55 5A 02 08 01 0B

Response:

55 5A 02 08 01 0B

Turn Off All Channels**Command:**

55 5A 02 0F 01 12

Response:

55 5A 02 0F 01 12

Query Channel Voltage (CMD: 0x03)**NOTE**

The device can measure the output voltage (VBUS) of each channel, which can be used to check if the bus is functioning normally.

The voltage unit is mV (millivolts).

Voltage resolution is 0.1 V

Data Field Description

data[0]	data[1]	data[2]
Channel ID	Channel Voltage value[15:8]	Channel Voltage value[7:0]
Channel_e	-	-

Query Channel 1 Voltage**Command:**

55 5A 03 01 00 04

Response:

55 5A 03 01 13 56 6D

Channel 1 Voltage = 0x1356 = 4950 mV (channel ON)**Query Channel 2 Voltage****Command:**

55 5A 03 02 00 05

Response:

55 5A 03 02 00 0C 11

Channel 2 Voltage = 0x000C = 12 mV (channel OFF)**Query Channel 3 Voltage****Command:**

55 5A 03 04 00 07

Response:

55 5A 03 04 00 09 10

Channel 3 Voltage = 0x0009 = 9 mV (channel OFF)**Query Channel 4 Voltage****Command:**

55 5A 03 08 00 0B

Response:

55 5A 03 08 00 08 13

Channel 4 Voltage = 0x0008 = 8 mV (channel OFF)**Query Channel Current (CMD: 0x04)****NOTE**

*The device can measure the current of each channel, which can be used to assess the device' s operating status.

*The current unit is mA (milliamps).

*Current resolution is 0.1 A

Data Field Description

data[0]	data[1]	data[2]
Channel ID	Channel Current value [15:8]	Channel Current value [7:0]
Channel_e	-	-

Query Channel 1 Current

Command:

```
55 5A 04 01 00 05
```

Response:

```
55 5A 04 01 01 29 2F
```

Channel 1 Current = 0x0129 = 297 mA

Query Channel 2 Current

Command:

```
55 5A 04 02 00 06
```

Response:

```
55 5A 04 02 00 00 06
```

Channel 2 Current = 0x0000 = 0 mA

Query Channel 3 Current

Command:

```
55 5A 04 04 00 08
```

Response:

```
55 5A 04 04 00 00 08
```

Channel 3 Current = 0x0000 = 0 mA

Query Channel 4 Current

Command:

```
55 5A 04 08 00 0C
```

Response:

```
55 5A 04 08 00 00 0C
```

Channel 4 Current = 0x0000 = 0ma

Set Button Control Permission (CMD: 0x09)



NOTE

- By default, the device allows channel control via the buttons.
- This setting applies to all buttons and channels.
- Long-press button functions are not affected.
- This setting is retained after power off.

Data Field Description

data[0]	data[1]
Fixed	Enable Button Control
0x00	0 (false), 1 (true)

Disable Control via Buttons

Command:

```
55 5A 09 00 00 09
```

Response:

```
55 5A 09 00 00 09
```

Enable Control via Buttons

Command:

```
55 5A 09 00 01 0A
```

Response:

```
55 5A 09 00 01 0A
```

Query Button Control Permission (CMD: 0x0A)

Data Field Description

data[0]	data[1]
Fixed	Enable Button Control
0x00	0 (false), 1 (true)

Command:

```
55 5A 0A 00 00 0A
```

Response:

Button control enabled:

```
55 5A 0A 00 01 0B
```

Button control disabled:

55 5A 0A 00 00 0A

Set Channel Power Default State (CMD: 0x0B)**NOTE**

When determining channel states after power-up, the default state has higher priority than the power-loss Persistence feature. If the default state feature is enabled, even if the power-loss Persistence feature is also enabled, the system on power-up will still use the default state to decide each channel's power or data on/off status.

When the default state feature is disabled, the channels' power-on state will be decided by the power-loss Persistence setting. If power-loss Persistence is also disabled (default configuration), then upon power-up all channels will default to OFF.

Factory default configuration: all channel power outputs are **OFF**.

Data Field Description

data[0]	data[1]	data[2]
Channel ID	Enable Default Power State	Default Power Value
Channel_e	0 (false), 1 (true)	0 (OFF), 1 (ON)

Set Channel 1 Power Default to ON**Command:**

55 5A 0B 01 01 01 0E

Response:

55 5A 0B 01 01 01 0E

Set Channel 1 Power Default to OFF**Command:**

55 5A 0B 01 01 00 0D

Response:

55 5A 0B 01 01 00 0D

Set Channel 2 Power Default to ON**Command:**

55 5A 0B 02 01 01 0F

Response:

55 5A 0B 02 01 01 0F

Set Channel 2 Power Default to OFF

Command:

```
55 5A 0B 02 01 00 0E
```

Response:

```
55 5A 0B 02 01 00 0E
```

Set Channel 3 Power Default to ON

Command:

```
55 5A 0B 04 01 01 11
```

Response:

```
55 5A 0B 04 01 01 11
```

Set Channel 3 Power Default to OFF

Command:

```
55 5A 0B 04 00 00 0F
```

Response:

```
55 5A 0B 04 00 00 0F
```

Set Channel 4 Power Default to ON

Command:

```
55 5A 0B 08 01 01 15
```

Response:

```
55 5A 0B 08 01 01 15
```

Set Channel 4 Power Default to OFF

Command:

```
55 5A 0B 08 01 00 14
```

Response:

```
55 5A 0B 08 01 00 14
```

Set All Channels Power Default to ON

Command:

```
55 5A 0B 0F 01 01 1C
```

Response:

```
55 5A 0B 0F 01 01 1C
```

Set All Channels Power Default to OFF

Command:

```
55 5A 0B 0F 01 00 1B
```

Response:

```
55 5A 0B 0F 01 00 1B
```

Disable Default Power State for Channel 1

Command:

```
55 5A 0B 01 00 00 0C
```

Response:

```
55 5A 0B 01 00 00 0C
```

Disable Default Power State for Channel 2

Command:

```
55 5A 0B 02 00 00 0D
```

Response:

```
55 5A 0B 02 00 00 0D
```

Disable Default Power State for Channel 3

Command:

```
55 5A 0B 04 00 00 0F
```

Response:

```
55 5A 0B 04 00 00 0F
```

Disable Default Power State for Channel 4

Command:

```
55 5A 0B 08 00 00 13
```

Response:

```
55 5A 0B 08 00 00 13
```

Disable Default Power State for All Channels

Command:

```
55 5A 0B 0F 00 00 1A
```

Response:

```
55 5A 0B 0F 00 00 1A
```

Query Channel Power Default State (CMD: 0x0C)

Data Field Description

data[0]	data[1]	data[2]
Channel ID	Enable Default Power State	Default Power Value
Channel_e	0 (false), 1 (true)	0 (OFF), 1 (ON)

Query Channel 1 Power Default State

Command:

```
55 5A 0C 01 00 00 0D
```

Response:

Default state disabled:

```
55 5A 0C 01 00 00 0D
```

Default OFF:

```
55 5A 0C 01 00 00 0D
```

Default ON:

```
55 5A 0C 01 01 01 0F
```

Query Channel 2 Power Default State

Command:

```
55 5A 0C 02 00 00 0E
```

Response:

Default state disabled:

```
55 5A 0C 02 00 00 0E
```

Default OFF:

```
55 5A 0C 02 00 00 0E
```

Default ON:

55 5A 0C 02 01 01 10

Query Channel 3 Power Default State

Command:

55 5A 0C 04 00 00 10

Response:

Default state disabled:

55 5A 0C 04 00 00 10

Default OFF:

55 5A 0C 04 00 00 10

Default ON:

55 5A 0C 04 01 01 12

Query Channel 4 Power Default State

Command:

55 5A 0C 08 00 00 14

Response:

Default state disabled:

55 5A 0C 08 00 00 14

Default OFF:

55 5A 0C 08 00 00 14

Default ON:

55 5A 0C 08 01 01 16

Query All Channels Power Default State

Command:

55 5A 0C 0F 00 00 1B

Response:

```
55 5A 0C 01 01 01 0F
55 5A 0C 02 01 01 10
55 5A 0C 04 01 01 12
55 5A 0C 08 01 01 16
```

- **Channel 1 Power: Default ON**
- **Channel 2 Power: Default ON**
- **Channel 3 Power: Default ON**
- **Channel 4 Power: Default ON**

Set Default Data Connectivity State (CMD: 0x0D)



NOTE

When determining channel states after power-up, the default state has higher priority than the power-loss Persistence feature. If the default state feature is enabled, even if the power-loss Persistence feature is also enabled, the system on power-up will still use the default state to decide each channel's power or data on/off status.

When the default state feature is disabled, the channels' power-on state will be decided by the power-loss Persistence setting. If power-loss Persistence is also disabled (default configuration), then upon power-up all channels will default to OFF.

Factory default configuration: all channels' data lines are connected by default.

Data Field Description

data[0]	data[1]	data[2]
Channel ID	Enable Default Data Line State	Default Data Line Value
Channel_e	0 (false), 1 (true)	0 (disconnect), 1 (connect)

Set Channel 1 Data Default to Connected

Command:

```
55 5A 0D 01 01 01 10
```

Response:

```
55 5A 0D 01 01 01 10
```

Set Channel 1 Data Default to Disconnected

Command:

```
55 5A 0D 01 01 00 0F
```

Response:

```
55 5A 0D 01 01 00 0F
```


Set Channel 2 Data Default to Connected

Command:

```
55 5A 0D 02 01 01 11
```

Response:

```
55 5A 0D 02 01 01 11
```

Set Channel 2 Data Default to Disconnected

Command:

```
55 5A 0D 02 01 00 10
```

Response:

```
55 5A 0D 02 01 00 10
```

Set Channel 3 Data Default to Connected

Command:

```
55 5A 0D 04 01 01 13
```

Response:

```
55 5A 0D 04 01 01 13
```

Set Channel 3 Data Default to Disconnected

Command:

```
55 5A 0D 04 01 00 12
```

Response:

```
55 5A 0D 04 01 00 12
```

Set Channel 4 Data Default to Connected

Command:

```
55 5A 0D 08 01 01 17
```

Response:

```
55 5A 0D 08 01 01 17
```

Set Channel 4 Data Default to Disconnected

Command:

```
55 5A 0D 08 01 00 16
```

Response:

```
55 5A 0D 08 01 00 16
```

Set All Channels Data Default to Connected

Command:

```
55 5A 0D 0F 01 01 1E
```

Response:

```
55 5A 0D 0F 01 01 1E
```

Set All Channels Data Default to Disconnected

Command:

```
55 5A 0D 0F 01 00 1D
```

Response:

```
55 5A 0D 0F 01 00 1D
```

Disable Default Data State for Channel 1

Command:

```
55 5A 0D 01 00 00 0E
```

Response:

```
55 5A 0D 01 00 00 0E
```

Disable Default Data State for Channel 2

Command:

```
55 5A 0D 02 00 00 0F
```

Response:

```
55 5A 0D 02 00 00 0F
```

Disable Default Data State for Channel 3

Command:

```
55 5A 0D 04 00 00 11
```

Response:

```
55 5A 0D 04 00 00 11
```

Disable Default Data State for Channel 4

Command:

```
55 5A 0D 08 00 00 15
```

Response:

```
55 5A 0D 08 00 00 15
```

Disable Default Data State for All Channels

Command:

```
55 5A 0D 0F 00 01 1D
```

Response:

```
55 5A 0D 0F 00 01 1D
```

Query Default Data Connectivity State (CMD: 0x0E)

Data Field Description

data[0]	data[1]	data[2]
Channel ID	Enable Default Data Connection State	Default Connection Value
Channel_e	0 (false), 1 (true)	0 (disconnect), 1 (connect)

Query Channel 1 Data Default State

Command:

```
55 5A 0E 01 00 00 0F
```

Response:

Default disabled, data line default is connected:

```
55 5A 0E 01 00 01 10
```

Default disabled, data line default is connected:

```
55 5A 0E 01 01 00 10
```

Default enabled, data line default is connected:

```
55 5A 0E 01 01 01 11
```

Query Channel 2 Data Default State

Command:

```
55 5A 0E 02 00 00 10
```

Response:

Default disabled, data line default is connected:

```
55 5A 0E 02 00 01 11
```

Default disabled, data line default is connected:

```
55 5A 0E 02 01 00 11
```

Default enabled, data line default is connected:

```
55 5A 0E 02 01 01 12
```

Query Channel 3 Data Default State

Command:

```
55 5A 0E 04 00 00 12
```

Response:

Default disabled, data line default is connected:

```
55 5A 0E 04 00 01 13
```

Default disabled, data line default is connected:

```
55 5A 0E 04 01 00 13
```

Default enabled, data line default is connected:

```
55 5A 0E 04 01 01 14
```

Query Channel 4 Data Default State

Command:

```
55 5A 0E 08 00 00 16
```

Response:

Default disabled, data line default is connected.

```
55 5A 0E 08 00 01 17
```

Default disabled, data line default is connected:

```
55 5A 0E 08 01 00 17
```

Default enabled, data line default is connected:

```
55 5A 0E 08 01 01 18
```

Query All Channels data line Default State

Command:

```
55 5A 0E 0F 00 00 1D
```

Response (scenario 1):

```
55 5A 0E 01 00 01 10
55 5A 0E 02 00 01 11
55 5A 0E 04 00 01 13
55 5A 0E 08 00 01 17
```

- Channel 1 data line: Default disabled, data connected
- Channel 2 data line: Default disabled, data connected
- Channel 3 data line: Default disabled, data connected
- Channel 4 data line: Default disabled, data connected

Response (scenario 2):

```
55 5A 0E 01 01 00 10
55 5A 0E 02 01 00 11
55 5A 0E 04 01 00 13
55 5A 0E 08 01 00 17
```

- Channel 1 data line: Default enabled, data line disconnected
- Channel 2 data line: Default enabled, data line disconnected
- Channel 3 data line: Default enabled, data line disconnected
- Channel 4 data line: Default enabled, data line disconnected

Set Power-Loss State Persistence (CMD: 0x0F)



NOTE: This feature can be toggled by long-pressing [Button \[2\]](#)

When enabled, the current channel states will be saved, and upon re-powering, all channels will restore to their state before power loss.

Data Field Description

data[0]	data[1]
Fixed	Enable Power-Off Persistence
0x00	0 (false), 1 (true)

Enable Power-Loss State Persistence

Command:

```
55 5A 0F 00 01 10
```

Response:

55 5A 0F 00 01 10

Disable Power-Loss State Persistence**Command:**

55 5A 0F 00 00 0F

Response:

55 5A 0F 00 00 0F

Query Power-Loss State Persistence (CMD: 0x10)**Data Field Description**

data[0]	data[1]
Fixed	Enable Power-Off Persistence
0x00	0 (false), 1 (true)

Command:

55 5A 10 00 00 10

Response:

- **Power-loss Persistence enabled:**

55 5A 10 00 01 11

- **Power-loss Persistence disabled:**

55 5A 10 00 00 10

Set Device Address (CMD: 0x11)**NOTE: 1. Assign a unique device address to each hub (this only needs to be done once)**

The device address is used to identify and distinguish each hub when multiple hubs are connected.

The address is user-defined and must be in the range: 0x0000 - 0xFFFF.

The default address is 0x0000.

Supported Products: All products (including SmartUSBHub Pro and FlexConnect) support the device address feature.

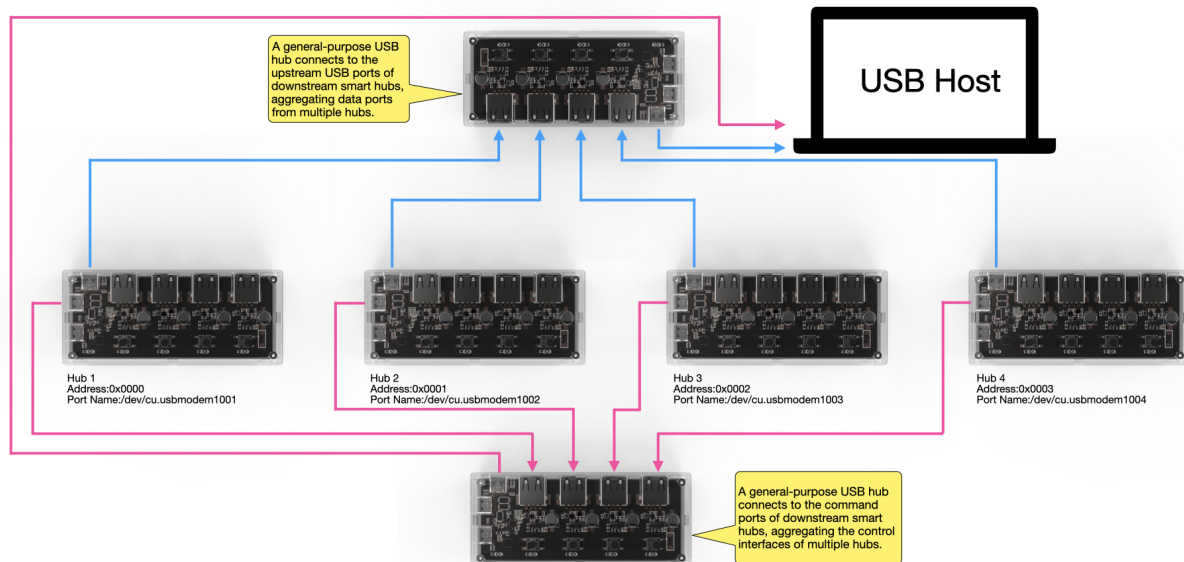
Usage:

- 1 In your program, sequentially read the device addresses of all connected hubs and bind them to their respective serial ports or instances.
- 2 After completing the above steps, each hub will have a clear and consistent index and mapping.

Connection Topology**Topology Structure 1**

- The upstream data ports of multiple hubs are connected to a higher-level hub.
- The command ports of multiple hubs are connected to a higher-level hub.
- Two top-level hubs are connected to the host.

The number of available downstream ports in this topology is $4n$.

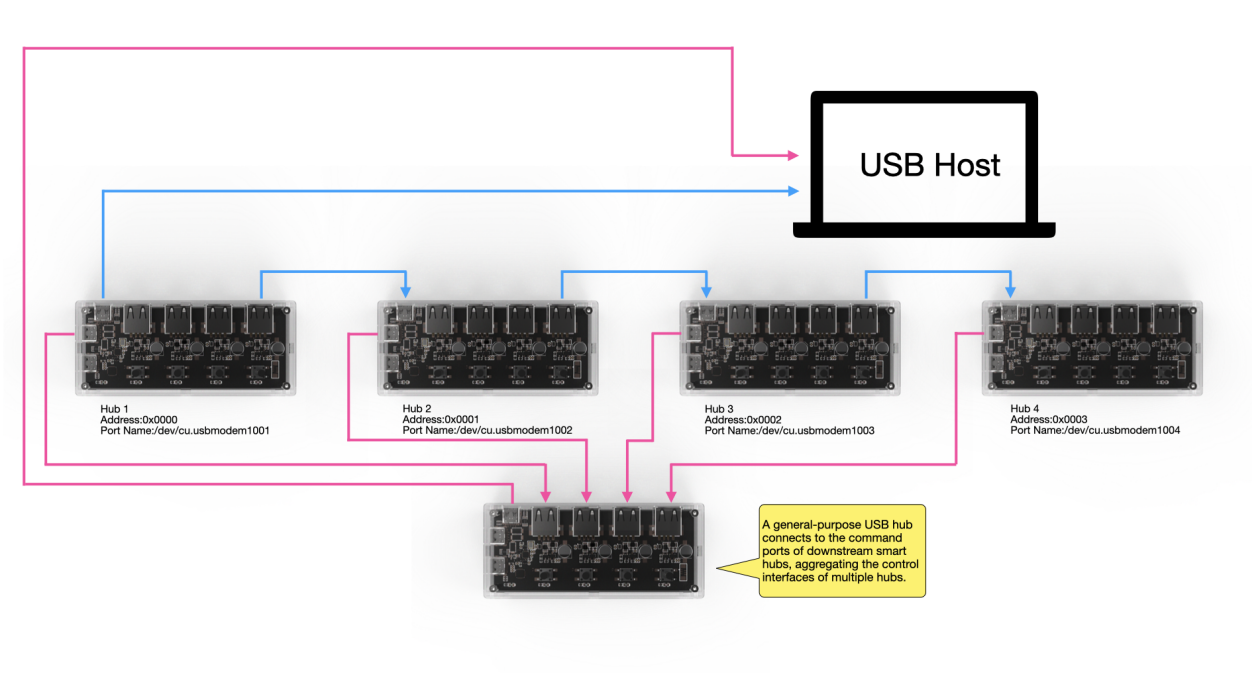


Topology.001

Topology Structure 2

- The upstream data port of the next hub is connected to the downstream data port of the previous hub.
- The command ports of multiple hubs are connected to a higher-level hub.
- The upstream data port of the first hub is connected to the host.
- The top-level hub in the control chain is connected to the host.

The number of available downstream ports in this topology is $3n+1$.



Topology.002

Data Field Description

data[0]	data[1]
Address [15–8]	Address [7–0]

Example

NOTE
Set the hubs device address to 0x0001

Command:
55 5A 11 00 01 12

Response:
55 5A 11 00 01 12

Get Device Address (CMD: 0x12)

Data Field Description

data[0]	data[1]
Address [15–8]	Address [7–0]
-	-

Example



NOTE

Get the hub's device address

Command:

55 5A 12 00 00 12

Response:

55 5A 12 00 01 13

The device address of this hub is: 0x0001

Set Operation Mode (CMD: 0x06)



NOTE: 1. Normal Mode: Each channel can be controlled independently

The device has two operation modes:

1Interlock Mode: Channels are mutually exclusive; only one channel can be on at a time.

This setting is retained after power off.

It can be toggled by long-pressing **Button 1** (>3 s). The LED indicator pattern reflects the current mode.

LED Indicator patterns:

Blink Pattern	Meaning
Normal operating mode	0.25 s on, 0.25 s off
Channel interlock mode	Momentary off flash (overlaid)
Receiving a control command frame	

Data Field Description

data[0]	data[1]
Fixed	Operating Mode
0x00	0 (Normal Mode), 1 (Interlock Mode)

Set to Normal Mode

Command:

55 5A 06 00 00 06

Response:

55 5A 06 00 00 06

Set to Interlock Mode

Command:

55 5A 06 00 01 07

Response:

55 5A 06 00 01 07

Query Operation Mode (CMD: 0x07)

Data Field Description

data[0]	data[1]
Fixed	Operating Mode
0x00	0 (Normal Mode), 1 (Interlock Mode)

Command:

```
55 5A 07 00 00 07
```

Response:

- **Device in Normal Mode:**

```
55 5A 07 00 00 07
```

- **Device in Interlock Mode:**

```
55 5A 07 00 01 08
```

Reset Factory Settings (CMD: 0xFC)

Data Field Description

data[0]	data[1]
Fixed	Fixed
0x00	0x00

Command:

```
55 5A FC 00 00 FC
```

Response:

```
55 5A FC 00 00 FC
```

Query Firmware Version (CMD: 0xFD)

Data Field Description

data[0]	data[1]
Version [15–8]	Version [7–0]
-	-

Command:

```
55 5A FD 00 00 FD
```

Response:

55 5A FD 00 0F 0C

Firmware Version: 15

Query Hardware Version (CMD: 0xFE)

Data Field Description

data[0]	data[1]
Version [15-8]	Version [7-0]
-	-

Command:

55 5A FE 00 00 FE

Response:

55 5A FE 00 03 01

Hardware Version: 3 / V1.3